
Hidden-Feature Matching and Left-Right Movement in Gait Representations

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Abstract

1 A representation of an evolving system should support estimates of observable
2 movement as well as its training task. We examine this requirement in gait using
3 a small skeleton Joint-Embedding Predictive Architecture (JEPA). The model pre-
4 dicts hidden skeleton features; a separate regression estimates a signed contrast
5 between left and right movement speeds from its frozen encoder. In source-video
6 cross-validation on 625 clips from 93 videos, trained predictors more consistently
7 match hidden features from the correct clip than from another video. Yet with an
8 expanded feature summary, mean source-balanced R^2 for the movement estimate
9 falls from 0.22 at initialization to 0.10–0.11 after training. Motion-weighted and
10 connected-region masks do not establish an improvement over their own random
11 references. This focused negative finding connects latent prediction to evaluation
12 of observable movement: success on hidden-feature matching did not improve
13 the specified movement estimate under the tested training and readout procedure.
14 Future-only prediction remains a separate test.

15 1 Introduction

16 Walking is an evolving system: coupled body landmarks change position over time, and movement
17 depends on their trajectories. Gait rhythm and coordination motivate temporal modeling, although
18 video-derived coordinates describe only part of the physical process. Temporal world modeling
19 aims to represent such a system well enough to predict its observable behavior. We test access to
20 one movement measurement within that larger objective.

21 We study that question with S-JEPA [1], using a left–right contrast of median coordinate speeds.
22 Clinical gait asymmetry motivates the setting [2]; our contrast has no clinical validation. Our central
23 null is no improvement from JEPA training over matched initial weights for this readout on held-
24 out source videos. The no-improvement null also motivated the registered reflection study. The
25 expanded readout and mask questions developed on the same cohort and remain exploratory.

26 The contribution is a domain-specific evaluation relevant to temporal foundation models: success on
27 latent prediction needs to be considered alongside access to observable movement. Scaling, broad
28 transfer, persistent temporal state and gait simulation remain outside the demonstrated scope.

29 2 Study design

30 **Data and observable.** Quality checks retain 625 GAVD clips from 93 source videos [3]. Each
31 archive has shape $T_i \times 33 \times 4$: three estimated coordinates plus landmark visibility, with original
32 frame numbers and frame rate. Coordinate units are image normalized with inferred depth; they are
33 not calibrated physical speeds.

34 The movement path uses observed coordinates, pelvis centering and a body-width scale, without
 35 gap filling or resizing. For five pairs—shoulders, knees, ankles, heels and foot tips—we calculate
 36 median speeds $m_{L,k}, m_{R,k}$ using original timestamp differences. Both sides must be observed at
 37 both ends of at least eight common transitions for every pair. The signed target is

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

38 Anatomical reflection exchanges left/right landmarks and validity flags and flips the centered hor-
 39 izontal coordinate, giving $y(Mx) = -y(x)$. Hips define the pelvis and are excluded from the
 40 contrast because centering equalizes their speed magnitudes. The statistic does not measure phase,
 41 stride timing or loading; reversing a trajectory together with its sampling intervals preserves its speed
 42 magnitudes.

43 **Encoder input.** A landmark is observed when visibility is at least 0.45 and all coordinates are finite.
 44 A separate encoder path fills interior gaps of at most four samples, treating filled positions as valid.
 45 After pelvis centering and body-width normalization, coordinates and validity are linearly resized to
 46 64 positions by sequence index. Original time intervals are discarded. Resized validity must reach
 47 0.999; invalid coordinates are zero. Each joint’s four-position patch projects 12 coordinates to 96
 48 features, giving a 16×33 grid. A patch is valid only if all four positions are valid. Anatomical
 49 identities and prepared positions survive this operation; the original trajectory, duration and cadence
 50 need not.

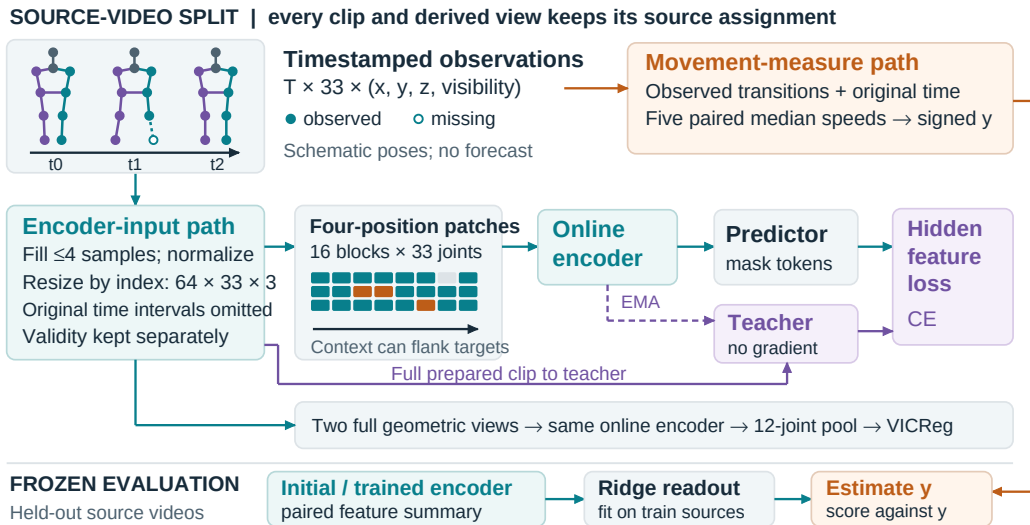


Figure 1: Two measurement paths and the completed JEPA training grid. Poses and mask cells are schematic. The movement target reaches only the separate regression evaluation. The online encoder has visible context on either side of a hidden position; the teacher and full-view regularizer access complete prepared clips.

51 **Training.** The grid uses 1,200 AdamW updates per encoder (architecture and settings in Appendix
 52 A). The online encoder zeroes hidden patch content before adding joint and position embeddings;
 53 the predictor substitutes learned mask tokens at target positions. Cross-entropy matches centered
 54 teacher feature distributions at hidden valid positions, averaged within clips and then across clips.
 55 A teacher follows the online weights by exponential moving average without gradients. A full-view
 56 VICReg term [4], weighted 0.05, pools twelve gait joints across two geometric views. Thus the total
 57 objective also accesses complete clips. Training uses CUDA BF16 and evaluation FP32.

58 We compare three motion conditions and two region conditions across five folds and five seeds: 125
 59 encoders. The motion conditions use random sampling, a MAMP-style sampler [5], and a robust
 60 motion mixture. Motion masks sample across all 33 joints using a count derived from the twelve-
 61 joint gait pool. Connected regions hide six connected landmarks over eight of sixteen blocks, with a

separate count-matched random reference. Each comparison matches initial weights, source draws, views and update counts. No reflection augmentation or reflection loss appears in this grid.

Held-out readout. Five outer folds keep each video’s clips and derived views together. Fresh encoders fit only outer-training videos with source-balanced sampling. All overlapping windows, mirrors and augmentations inherit the source split. The encoder is frozen while ridge regression, a linear model with a coefficient penalty, estimates y . Three inner source folds select the penalty using only outer-training labels; imputation and scaling are fitted within each inner training set and then refitted with the regression on all outer-training sources. Inner-validation clips may have contributed without labels to outer-training encoder fitting.

Both encoders use bilateral mean features (960 inputs), expanded with standard deviations, absolute adjacent-block feature changes and ten validity-support fractions (2,890 inputs). We pool outer-test predictions within each seed, weighting each video equally, then average the five R^2 values. Paired 95% source-bootstrap intervals resample videos with their clips and seed predictions together (2,000 resamples). They condition on the fitted models and existing splits, excluding uncertainty from retraining, alternative partitions and development choices. Repeated seeds and mask draws do not add independent videos.

3 Findings

Movement readout falls after training. The expanded initial representation reaches mean $R^2 = 0.223$, compared with 0.101–0.114 for trained teachers (Figure 2). All five conditions decrease in every seed, with greater mean absolute error. Paired trained-minus-initial intervals are below zero in every condition. Final online encoders also score below their matched initial encoders.

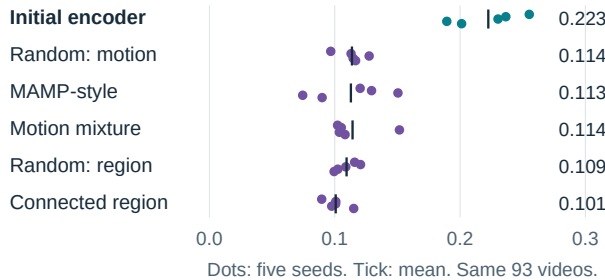
Training mask	Mean R^2	ΔR^2	95% source interval
Random (motion)	0.114	-0.109	[-0.170, -0.041]
MAMP-style	0.113	-0.110	[-0.165, -0.051]
Motion mixture	0.114	-0.108	[-0.171, -0.042]
Random (region)	0.109	-0.113	[-0.166, -0.060]
Connected region	0.101	-0.122	[-0.183, -0.055]

Table 1: trained teacher minus matched initialization. Intervals resample source videos, conditional on fitted models; differences are calculated before rounding.

Expanding the initial readout raises R^2 from 0.071 to 0.223 (+0.152). The added components mix variation, feature changes and observation support, so the gain cannot be assigned specifically to temporal order. The initial control retains the pretrained pose detector, input preparation, anatomical identities and supervised regression.

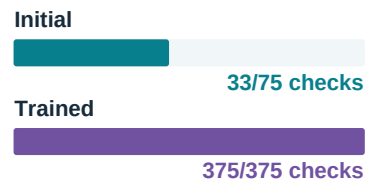
A Movement estimate from frozen features

Expanded summary; higher source-balanced R^2 is better



B Hidden-feature matching

Correct clip beats another source



Counts summarize repeated model checks; they are not independent experiments. Neither panel evaluates future movement.

Figure 2: Different outcomes from the same fitted models. A: frozen movement readout; points are five seed scores, and short vertical marks their mean. B: correct-clip feature error is lower in 375 of 375 trained diagnostic checks, versus 33 of 75 distinct initial checks. These checks reuse models and videos.

90 **Correct-clip features become easier to match.** Each diagnostic compares source-weighted mean
91 squared feature error against the correct clip and valid targets from another source, holding the eval-
92 uated clips and hidden positions fixed. All 375 trained checks favor correct targets; 33 of 75 initial
93 checks do so. The trained checks cover 125 fitted models under three evaluation masks (left-leg gap,
94 right-leg gap and scattered gap). Initial checks cover 25 matched fold/seed controls under the same
95 three masks, with copies shared between experiment families counted once. This preference could
96 involve pose, viewpoint or missingness as well as motion. Teachers and target distributions change
97 during training, so raw training losses cannot rank representation quality across conditions.

98 **Changing mask geometry does not establish better readout.** The inspection draws confirm that
99 motion samplers select higher-motion targets and that connected regions remove immediate tempo-
100 ral brackets. Motion masks hide approximately 17% of valid all-landmark tokens; regions hide 9.9%.
101 Their readout comparisons therefore use distinct random references. All three mask-comparison
102 intervals span zero (Appendix B). Their benefits remain uncertain; the intervals do not establish
103 equivalence.

104 **Preparing input changes the measurement.** A saved comparison recalculates the target after input
105 preparation: 623 clips from 92 sources have finite values on both paths, with source-weighted sign
106 agreement 70.4% and calculation-agreement $R^2 = 0.218$. This is agreement between two calcula-
107 tions of the observable, not a learned model score. Neither the achievable readout accuracy nor an
108 individual preprocessing cause follows from it.

109 4 Interpretation and next experiment

110 The two evaluations answer different questions. JEPA training improves preference for the correct
111 clip’s hidden features, while a fixed regression procedure estimates its left–right speed contrast less
112 accurately. Poorer readout performance does not imply complete destruction of movement informa-
113 tion. In particular, 49/125 teacher fits and 96/125 online fits select the largest ridge penalty, 10,000.
114 Wider training-only selection, support-only controls and summaries of matched dimension would
115 help separate feature geometry from readout limitations. Source holdout does not establish partic-
116 ipant independence, chronological holdout or an untouched development cohort. Recording and
117 pose-estimation cues remain possible explanations.

118 A next experiment would decode future coordinates from past-only inputs at common valid end-
119 points and physical-time horizons. Fit and select decoders on training sources only; compare last-
120 position, constant-velocity, direct-pose, initial-encoder and mismatched-training-future controls. Fu-
121 ture coordinates and visibility must not change prefix preparation, normalization, masks or predic-
122 tions. The retained past-feature readout and future-feature decoding studies have synthetic training
123 evidence; the real-data horizon census establishes eligibility only. Decoding observed future-teacher
124 features diagnoses the decoder; decoding features predicted from the past evaluates forecasts. Mul-
125 tiple horizons from one prefix are not recursive rollouts.

126 The finding is conditional on this cohort, recipe and readout. Forecasting and simulation require
127 tests of future behavior beyond the whole-clip contrast.

128 5 Related work

129 Skeleton action recognition motivates learned feature targets [1], motion targets [5] and connected-
130 body masking [6]. We vary masks within one JEPA objective without reproducing these complete
131 methods. Video anticipation and planning [7], together with transformation-sensitive representa-
132 tion learning [8], show why evaluation depends on the intended behavior. Here the endpoint is an
133 observed gait quantity: comparing its recovery before and after training complements those task
134 evaluations.

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168 A Measurement and implementation details

169 Of 666 selected annotations, 642 match pose archives and 625 pass quality checks. The motion
170 target count is half the batch’s smallest valid twelve-joint gait pool, rounded down, then sampled
171 across all 33 joints.

172 The completed recipe uses 96 features, encoder/predictor depths 4/2, four attention heads, batch size
173 20, 1,200 AdamW updates, learning rate 0.001, weight decay 0.05 and teacher momentum 0.999.

174 The archive coordinates are $x = (x_0 + uW_c)/W$, $y = (y_0 + vH_c)/H$, and $z = dW_c/W$, where crop
175 origin and size are (x_0, y_0) , W_c , H_c , frame size is W , H , and u, v, d are pose estimates. Unequal
176 image-axis scaling and inferred depth remain after body normalization. The scale is the median of
177 valid shoulder and hip widths. The movement path requires observed hips; the encoder uses the
178 median valid pelvis midpoint for missing hips (zero if none is available). No boundary extrapolation
179 fills gaps.

180 The teacher receives the unaugmented prepared clip; the masked online path receives a geometric
181 view. Two full online views use rotations up to 8 degrees in the horizontal-depth plane and transla-
182 tions up to 0.03 in image-plane coordinates. Target/predictor temperatures are 0.06/0.10; the teacher
183 center uses momentum 0.9. Gradient norm is clipped at 1. The twelve-joint regularizer pool is
184 shoulders, hips, knees, ankles, heels and foot tips. Its projector is a separate training component.
185 It combines invariance, variance and covariance penalties with internal weights 25, 25 and 1, then
186 overall weight 0.05.

187 The expanded summary uses common left/right validity within each pair; increments require con-
188 secutive common valid blocks. Each pair contributes differences and sums of means, standard de-
189 viations and mean absolute feature increments (six 96-vectors), plus two support fractions. Thus
190 $5 \times 6 \times 96 + 10 = 2,890$. Missing support gives zero summary values. Temporal positions in
191 this summary are prepared block indices, not original timestamps. Ridge penalties are selected from
192 0.01, 0.1, 1, 10, 100, 1000, 10000.

193 Source-balanced R^2 uses weights $w_{vi} = 1/(Vn_v)$ and weighted test mean $\bar{y}_w$:

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

194 The test mean defines this metric’s denominator, not a deployable fitted baseline. Outer test folds
195 contain 19, 19, 19, 18 and 18 videos and 189, 182, 72, 77 and 105 clips. The retained provenance
196 audit checked archive IDs and hashes, source membership and absence of test-source training draws.
197 Those checks concern file integrity and split adherence, not cross-upload content deduplication. Per-
198 sistent person IDs and complete annotation retrieval/version records are unavailable, preventing a
199 person-level contamination audit or full reconstruction of acquisition history. Source holdout alone
200 provides no evidence of generalization under distribution shift.

201 B Supporting results and reproducibility

202 MAMP-style, motion-mixture and region masks minus their own random references give
203 $\Delta R^2 = -0.0008$, $+0.0005$ and -0.009 , with intervals $[-0.020, 0.015]$, $[-0.015, 0.015]$ and
204 $[-0.033, 0.018]$. A 264-value coordinate/validity summary reaches $R^2 = 0.035$; it is a limited
205 direct-pose baseline, not a ceiling on coordinate information.

206 These exploratory percentile intervals use bootstrap seed 812, retain all five fitted seeds and have no
207 familywise or development-selection adjustment. The numerical supplement preserves aggregate
208 seed scores and recorded intervals. The executed report identifies 125,000 prediction rows and 200
209 pooled score rows; a saved audit records their recomputation. Raw predictions and checkpoints
210 are unavailable in the revision environment, so this revision checks aggregate consistency without
211 claiming to repeat that bootstrap or inference.

212 A separate reflection-augmentation comparison has real-data summary evidence: augmentation re-
213 duces the specified teacher-token reflection discrepancy by $\Delta q = -0.008$ $[-0.010, -0.007]$, while
214 its readout difference is $\Delta R^2 = 0.004$ $[-0.006, 0.013]$. This uses its own recipe, four inner readout
215 folds and compact summary; it is not pooled with the motion grid. An explicit reflection loss has

216 synthetic evidence only. Likewise, a readout forced to take the form $g(x) = [h(x) - h(Mx)]/2$
217 reverses sign algebraically without establishing learned movement information.

218 GAVD annotations describe the cohort, not affected side or treatment response. Uncalibrated coordi-
219 nates and absent clinical reference measurements preclude a clinical interpretation of y . Public
220 annotation access does not authorize redistribution of separately hosted videos or derived poses;
221 neither is included in the supplement.

222 **AI assistance.** An AI assistant supported author-side evidence reconciliation, literature checking,
223 manuscript revision, figure programming and adversarial checks. No new model training was per-
224 formed for this revision. Responsibility for the submitted text, figures, claims and references remains
225 with the authors.